

Continuous Functions on Product Spaces

We begin this chapter with a proof that under suitable hypotheses a function f continuous on a product space X_I (or even on some of its dense subspaces) is effectively defined on a ‘small’ subproduct (in the sense that there are ‘small’ $J \subset I$ and continuous g on X_J such that $f = g \circ \pi_J$). We apply this result, together with some of the product-space theorems of Chapter 3, to obtain an identification in concrete form of the Stone–Čech compactification, the Hewitt realcompactification, and the Dieudonné topological completion of certain completely regular Hausdorff spaces (these terms are defined in Appendix B). Such an opportunity occurs infrequently since the Stone–Čech compactification of a space X , whose elements are normally described as maximal filters in the class of zero-sets of X , is defined by an appeal to (a variant of) Zorn’s lemma; it does not arise in practice as a readily identifiable, familiar space.

Among the specific results proved are these (cf. Corollary 10.7 and Theorem 10.17): in a product of metrizable spaces, and in a product of compact spaces, every ω^+ - Σ -space is C -embedded. We conclude with an example, one of several due to Ulmer, showing that there do exist ω^+ - Σ -products $\Sigma_{\omega^+}(p, X_I)$ not C -embedded in X_I .

Definition. Let $Y \subset X_I$, $J \subset I$, let Z be a set and $f: Y \rightarrow Z$. Then f depends on J (in Y) if $f(y) = f(y')$ whenever $y, y' \in Y$ and $y_J = y'_J$.

If α is a cardinal and there is $J \in \mathcal{P}_\alpha(I)$ such that f depends on J , then we say that f depends on fewer than α coordinates, or that f depends on $< \alpha$ coordinates; the expression f depends on $\leq \alpha$ coordinates is defined analogously.

We note that if f depends on J and $J \subset J' \subset I$ then f depends on J' .

We will show that under certain circumstances a (continuous) function will depend on a small set of coordinates. We note in passing

that for sufficiently ill-behaved sets Y it will occur that every function $f: Y \rightarrow Z$, regardless of the nature of Z and regardless of whether or not f is continuous, depends on any set J . For an example, let $X_\xi = \mathbb{R}$ for $\xi < 2^\omega$ and let $D = \{x(n) : n < \omega\}$ be a (countable) dense subset of $\prod_{\xi < 2^\omega} X_\xi$ such that

$$x(n)_\xi \neq x(n')_\xi \text{ whenever } \xi < 2^\omega, n, n' < \omega, n \neq n';$$

then for every set $Z, f: D \rightarrow Z$ and $J \subset 2^\omega$, the function f depends on J .

The following lemma, crucial to the results of this chapter, does not depend on the Erdős–Rado theorem.

10.1 Lemma. Let $\omega \leq \kappa \leq \alpha$ with either $\kappa < \alpha$ or α regular, let $\{X_i : i \in I\}$ be a set of non-empty spaces, Y a dense, pseudo- (α, κ) -compact subspace of $(X_I)_\kappa$, (M, ρ) a metric space and f a continuous function from Y to M . Then for every $\varepsilon > 0$ there is $J \in \mathcal{P}_\alpha(I)$ such that

$$\rho(f(x), f(y)) \leq \varepsilon \quad \text{if } x, y \in Y, x_J = y_J.$$

Proof. We suppose the result fails.

We claim that for $J \in \mathcal{P}_\alpha(I)$ there are $x, y \in Y$ such that $x_J = y_J$, $\rho(f(x), f(y)) > \varepsilon$, and

$$|\{i \in I : x_i \neq y_i\}| < \kappa.$$

Indeed, there are $\tilde{x}, \tilde{y} \in Y$ such that $\tilde{x}_J = \tilde{y}_J$ and $\rho(f(\tilde{x}), f(\tilde{y})) > \varepsilon$, and since f is continuous there are basic neighborhoods U and V of $\tilde{x}$ and $\tilde{y}$, respectively, such that

$$\rho(f(x), f(y)) > \varepsilon \quad \text{for } x \in U \cap Y, y \in V \cap Y.$$

We set $A = R(U) \cup R(V)$, and we define x, y by

$$\begin{aligned} x_i &= \tilde{x}_i & \text{if } i \in J \cup A, \\ y_i &= \tilde{y}_i & \text{if } i \in J \cup A, \\ x_i &= y_i = \tilde{x}_i & \text{if } i \in I \setminus (J \cup A). \end{aligned}$$

Then x, y are as required. The claim is established.

For $\xi < \alpha$ we define $x(\xi), y(\xi), U^\xi, V^\xi, J(\xi)$ and $A(\xi)$ so that

- (i) $x(\xi), y(\xi) \in Y$ for $\xi < \alpha$;
- (ii) $A(\xi) = \{i \in I : x(\xi)_i \neq y(\xi)_i\}$ for $\xi < \alpha$;
- (iii) $A(\xi) \in \mathcal{P}_\kappa(I)$ for $\xi < \alpha$;

- (iv) U^ξ and V^ξ are basic neighborhoods in $(X_I)_\kappa$ of $x(\xi)$ and $y(\xi)$, respectively, for $\xi < \alpha$;
- (v) $\rho(f(x), f(y)) > \varepsilon$ if $x \in U^\xi \cap Y$, $y \in V^\xi \cap Y$, for $\xi < \alpha$;
- (vi) $R(U^\xi) = R(V^\xi)$ for $\xi < \alpha$;
- (vii) $U_i^\xi \cap V_i^\xi = \emptyset$ if $i \in A(\xi)$, for $\xi < \alpha$;
- (viii) $J(0) = \emptyset$;
- (ix) $J(\xi) = (\cup_{\eta < \xi} A(\eta)) \cup (\cup_{\eta < \xi} R(U^\eta))$ for $0 < \xi < \alpha$; and
- (x) $x(\xi)_{J(\xi)} = y(\xi)_{J(\xi)}$ for $\xi < \alpha$.

We proceed by recursion. If $\xi < \alpha$ and $x(\eta)$, $y(\eta)$, U^η , V^η , $J(\eta)$ and $A(\eta)$ are defined for $\eta < \xi$, we define $J(\xi)$ by (viii) or (ix) and we use the claim just established to define $x(\xi)$, $y(\xi)$ and $A(\xi)$ satisfying (i), (x), (ii) and (iii). There are neighborhoods $\tilde{U}^\xi$ and $\tilde{V}^\xi$ of $x(\xi)$ and $y(\xi)$, respectively, such that

$$\rho(f(x), f(y)) > \varepsilon \quad \text{for } x \in \tilde{U}^\xi \cap Y, y \in \tilde{V}^\xi \cap Y.$$

We define neighborhoods U_i^ξ and V_i^ξ in X_i of $x(\xi)_i$ and $y(\xi)_i$, respectively so that

$$\begin{aligned} U_i^\xi \cap V_i^\xi &= \emptyset \quad \text{for } i \in A(\xi), \\ U_i^\xi \subset \tilde{U}_i^\xi \quad \text{and} \quad V_i^\xi \subset \tilde{V}_i^\xi \quad &\text{for } i \in A(\xi), \text{ and} \\ U_i^\xi = V_i^\xi = \tilde{U}_i^\xi \cap \tilde{V}_i^\xi \quad &\text{for } i \in I \setminus A(\xi); \end{aligned}$$

it is then clear that (iv), (v), (vi) and (vii) are satisfied.

Now for $i \in I$ set $B(i) = \{\xi < \alpha : U_i^\xi \neq V_i^\xi\}$. Since $\xi \in B(i)$ if and only if $i \in A(\xi)$, and since

$$A(\eta) \cap A(\xi) = \emptyset \quad \text{for } \eta < \xi < \alpha$$

by (ix), we have $|B(i)| \leq 1$ for $i \in I$.

For $p \in Y$, let $\mathcal{A}(p)$ denote the set of basic neighborhoods in $(X_I)_\kappa$ of p , and for $W \in \mathcal{A}(p)$ set

$$A(p, W) = \{\xi < \alpha : W \cap U^\xi \neq \emptyset\}.$$

Since Y is pseudo- (α, κ) -compact, there is $\bar{p} \in Y$ such that $|A(\bar{p}, W)| \geq \kappa$ for all $W \in \mathcal{A}(\bar{p})$. We claim that for $W \in \mathcal{A}(\bar{p})$ there is $\bar{\xi} \in A(\bar{p}, W)$ such that $W \cap V^{\bar{\xi}} \neq \emptyset$. Indeed, since $|A(\bar{p}, W)| \geq \kappa$ and $|R(W)| < \kappa$ and $|B(i)| \leq 1$ for all $i \in R(W)$, there is $\bar{\xi} \in A(p, W)$ such that $\bar{\xi} \notin B(i)$ for $i \in R(W)$; from $W \cap U^{\bar{\xi}} \neq \emptyset$ and $U_i^{\bar{\xi}} = V_i^{\bar{\xi}}$ for all $i \in R(W)$ it follows that $W \cap V^{\bar{\xi}} \neq \emptyset$. The claim is proved.

Since Y is dense in $(X_I)_\kappa$, we have

$$W \cap U^{\bar{\zeta}} \cap Y \neq \emptyset \text{ and } W \cap V^{\bar{\zeta}} \cap Y \neq \emptyset$$

for all $W \in \mathcal{A}(\bar{p})$. From (v) it follows that f is not continuous at $\bar{p}$. This contradiction completes the proof.

10.2 Corollary. Let $\omega \leq \kappa \leq \alpha$ with either $\kappa < \alpha$ or α regular, let $\{X_i : i \in I\}$ be a set of non-empty spaces, Y a dense, pseudo- (α, κ) -compact subspace of $(X_I)_\kappa$, (M, ρ) a metric space and f a continuous function from Y to M . Then

- (a) if $\text{cf}(\alpha) = \omega$ then f depends on $\leq \alpha$ coordinates;
- (b) if $\text{cf}(\alpha) > \omega$ then f depends on $< \alpha$ coordinates.

Proof. For $1 \leq n < \omega$ there is (by Lemma 5.1) $J_n \in \mathcal{P}_\alpha(I)$ such that

$$\rho(f(x), f(y)) \leq 1/n \quad \text{if } x, y \in Y, x_{J_n} = y_{J_n}.$$

We set $J = \cup \{J_n : 1 \leq n < \omega\}$. Then f depends on J , and

$$\begin{aligned} |J| &\leq \alpha \quad \text{in (a)} \\ &< \alpha \quad \text{in (b).} \end{aligned}$$

10.3 Lemma. Let $\alpha \geq \kappa \geq \omega$, let $\{X_i : i \in I\}$ be a set of non-empty spaces, Y a subspace of $(X_I)_\kappa$ such that $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| < \alpha$, Z a space and f a continuous function from Y to Z such that f depends on $< \alpha$ coordinates. Then there is continuous $\bar{f} : (X_I)_\kappa \rightarrow Z$ such that $f \subset \bar{f}$.

Proof. There is $J \in \mathcal{P}^*(I)$ such that f depends on J . For $x \in X_I$ we choose $\bar{x} \in Y$ such that $x_J = \bar{x}_J$ (such $\bar{x}$ exists because $x_J \in X_J = \pi_J[Y]$), and we set

$$\bar{f}(x) = f(\bar{x}).$$

For $y \in Y$ we have $y_J = \bar{y}_J$ and hence

$$\bar{f}(y) = f(\bar{y}) = f(y)$$

(since f depends on J); thus $f \subset \bar{f}$. It remains to show that $\bar{f}$ is continuous.

Let $x \in X$ and W a neighborhood in Z of $\bar{f}(x)$ and let U and V be neighborhoods of $\bar{x}_J$ and $\bar{x}_{I \setminus J}$ in $(X_J)_\kappa$ and $(X_{I \setminus J})_\kappa$, respectively, such that

$$f[(U \times V) \cap Y] \subset W.$$

Since $x_j = \bar{x}_j \in U$, the set $\pi_j^{-1}(U)$ is a neighborhood in $(X_I)_\kappa$ of x . We claim that if $x' \in \pi_j^{-1}(U)$ then $\tilde{f}(x') \in W$. Indeed, since $|J \cup R(V)| < \alpha$ there is $y' \in Y$ such that

$$\begin{aligned} y'_i &= x'_i & \text{if } i \in J \\ &\in V_i & \text{if } i \in R(V) \setminus J \\ &\in X_i & \text{if } i \in I \setminus (J \cup R(V)), \end{aligned}$$

and from $y' \in (U \times V) \cap Y$ we have

$$\tilde{f}(x') = f(\overline{x'}) = f(y') \in W,$$

as required.

The proof is complete.

Even in the case $\kappa = \omega$ and $Z = [0, 1]$ of Lemma 10.3 it is not known whether the hypothesis that $\pi_J[Y] = X_J$ for all $J \in \mathcal{P}_\alpha^*(I)$ can be replaced by the weaker hypothesis that Y is dense in $(X_J)_\kappa$ and (for all $J \in \mathcal{P}_\alpha^*(I)$) every continuous $f: \pi_J[Y] \rightarrow Z$ extends to continuous $\tilde{f}: (X_J)_\kappa \rightarrow Z$.

10.4 Theorem. Let α and κ be infinite cardinals, $\{X_i: i \in I\}$ a set of spaces, and Y a subspace of $(X_I)_\kappa$ such that $\pi_J[Y] = (X_J)_\kappa$ for every non-empty $J \subset I$ with $|J| < \alpha$. Assume that either

(i) $\omega \leq \kappa \ll \alpha$ with α regular and $(X_J)_\kappa$ is pseudo- (α, κ) -compact for every non-empty $J \subset I$ with $|J| < \kappa$, or

(ii) $\omega \leq \kappa \ll \alpha$, $\kappa \ll \text{cf}(\alpha)$ and $(X_J)_\kappa$ is pseudo- (α, α) -compact for every non-empty $J \subset I$ with $|J| < \kappa$.

Then Y is M -embedded in $(X_I)_\kappa$.

Proof. It follows in (i) from Theorem 3.5(c) that Y is pseudo- (α, κ) -compact, and it follows in (ii) from Theorem 3.13 that Y is pseudo- (α, α) -compact. Thus Y is a dense, pseudo- (α, κ) -compact subspace of $(X_I)_\kappa$, and from Corollary 10.2 we have that every continuous function from Y to a metric space depends on $< \alpha$ coordinates. That Y is M -embedded in $(X_I)_\kappa$ then follows from Lemma 10.3.

10.5 Corollary. Let α be an infinite cardinal, $\{X_i: i \in I\}$ a set of spaces, and Y a subspace of X_I such that $\pi_J[Y] = X_J$ for all non-empty $J \subset I$ with $|J| < \alpha$. Assume that either

(i) $\omega < \alpha$ with α regular, X_i is a regular Hausdorff space for $i \in I$ and $S(X_J) \leq \alpha$ for all non-empty $J \subset I$ with $|J| < \omega$, or

(ii) $\omega < \text{cf}(\alpha)$ and $d(X_i) < \text{cf}(\alpha)$ for $i \in I$.

Then Y is M -embedded in X_I .

Proof. In (i) for $J \in \mathcal{P}_\omega^*(I)$ the space X_J has calibre $(\alpha, 2)$, hence by Theorem 2.1(e) is pseudo- $(\alpha, 2)$ -compact, hence by Theorem 2.1(i) is pseudo- (α, ω) -compact.

In (ii) for $J \in \mathcal{P}_\omega^*(I)$ we have $d(X_J) < \text{cf}(\alpha)$ and therefore by Theorem 2.7 the space X_J has calibre (α, α) ; thus by Theorem 2.1(e) for $J \in \mathcal{P}_\omega^*(I)$ the space X_J is pseudo- (α, α) -compact.

The required conclusion now follows from (the case $\kappa = \omega$ of) Theorem 10.4.

It follows from the uniqueness of the Dieudonné topological completion of a completely regular, Hausdorff space (cf. Theorem B.2 of the Appendix) that if Y and $(X_I)_\kappa$ are related as in Theorem 10.4 then $\gamma(Y) = \gamma((X_I)_\kappa)$. This makes it possible to identify specifically and in concrete form the Hewitt realcompactification and the Stone–Čech compactification of certain spaces (cf. Corollary 10.7 below).

For $\alpha \geq \omega$ we say that a space X is *weakly α -compact* if for every cover $\{V_i : i \in I\}$ of X by open sets there is $J \subset I$ with $|J| < \alpha$ such that $\cup_{i \in I} V_i$ is dense in X .

10.6 Lemma. Let $\alpha \geq \omega$ and let X be a space.

(a) If $S(X) \leq \alpha$, then X is weakly α -compact; and

(b) if X is weakly α -compact, then X is pseudo- α -compact.

Proof. (a) Let $\{V_i : i \in I\}$ be a cover of X by open sets and let $\{U_j : j \in J\}$ be a family of non-empty open subsets of X such that

(i) $U_j \cap U_{j'} = \emptyset$ for $j, j' \in J, j \neq j'$ (i.e., the indexed family $\{U_j : j \in J\}$ is a cellular family),

(ii) for $j \in J$ there is $i \in I$ such that $U_j \subset V_i$, and

(iii) the family $\{U_j : j \in J\}$ is maximal with respect to (i) and (ii).

It follows from (iii) that $\cup_{j \in J} U_j$ is dense in X , and from (i) that $|J| < \alpha$. For $j \in J$ we choose $i(j) \in I$ such that $U_j \subset V_{i(j)}$, and from

$$\cup_{j \in J} U_j \subset \cup_{j \in J} V_{i(j)} \subset X$$

we note that $\cup_{j \in J} V_{i(j)}$ is dense in X , as required.

(b) Let $\{U_\xi : \xi < \beta\}$ be a locally finite (indexed) family of non-empty open subsets of X ; we show $\beta < \alpha$. There is an open cover $\{V_i : i \in I\}$ of X such that $|\{\xi < \beta : V_i \cap U_\xi \neq \emptyset\}| < \omega$ for $i \in I$. Since X is weakly α -compact, there is $J \subset I$ with $|J| < \alpha$ such that $\cup_{i \in J} V_i$ is dense in X . We set

$$B(i) = \{\xi < \beta : V_i \cap U_\xi \neq \emptyset\} \text{ for } i \in J,$$

and we note that since $\cup_{i \in J} V_i$ is dense in X we have $\cup_{i \in J} B(i) = \beta$. From $|B(i)| < \omega$ and $|J| < \alpha$ it now follows that $\beta < \alpha$, as required.

10.7 Corollary. Let $\{X_i : i \in I\}$ be a set of realcompact spaces and Y a subspace of X_I such that $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \omega$.

(a) If either

- (i) $d(X_i) \leq \omega$ for all $i \in I$, or
- (ii) $S(X_J) \leq \omega^+$ for every non-empty $J \subset I$ with $|J| < \omega$, or
- (iii) X_J is a Lindelöf space for every non-empty $J \subset I$ with $|J| < \omega$, or
- (iv) X_J is weakly ω^+ -compact for every non-empty $J \subset I$ with $|J| < \omega$, or
- (v) X_J is pseudo- ω^+ -compact for every non-empty $J \subset I$ with $|J| < \omega$,

then $X_I = v(Y)$.

(b) If each space X_i is compact, then $X_I = \beta(Y)$.

Proof. (a) Properties (i) through (v) are related by the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iv) $\Rightarrow$ (v) and (iii) $\Rightarrow$ (iv), so it is sufficient to show from (v) that $X_I = v(Y)$. From (the case $\kappa = \omega$, $\alpha = \omega^+$ of) Theorem 10.4(i) it follows that Y is M -embedded (and *a fortiori* C -embedded) in X_I . Since X_I is realcompact, we have

$$v(Y) = v(X_I) = X_I$$

from Theorem B.4 of the Appendix.

(b) Again from Theorem 10.4(i) it follows that Y is M -embedded (and *a fortiori* C^* -embedded) in X_I . Since X_I is compact we have

$$\beta(Y) = \beta(X_I) = X_I$$

from Theorem B.4 of the Appendix.

We continue with another consequence of Theorem 10.4: under

appropriate (and quite stringent) conditions, homeomorphic subspaces of a product space have homeomorphic complements.

10.8 Theorem. Let $\omega < \text{cf}(\alpha)$, let $\{X_i : i \in I\}$ be a set of non-compact metric spaces with $|I| \geq \alpha$ and with $d(X_i) < \text{cf}(\alpha)$ for $i \in I$, and let A, A' be subsets of X_I such that either

$$|A \cup A'| < 2^{|I|}, \text{ or}$$

there is a set $\{K_\xi : \xi < \beta\}$ of compact subspaces of X_I with $\beta \leq \alpha$ such that $A \cup A' \subset \bigcup_{\xi < \beta} K_\xi$.

Then if $X_I \setminus A$ and $X_I \setminus A'$ are homeomorphic, then A and A' are homeomorphic.

Proof. We set $Y = X_I \setminus (A \cup A')$ and we claim that if $J \in \mathcal{P}_\alpha^*(I)$ then $\pi_J[Y] = X_J$. If the claim fails there is $p \in X_J \setminus \pi_J[Y]$ and we have $\{p\} \times X_{I \setminus J} \subset A \cup A'$. We consider two cases.

Case 1. $|A \cup A'| < 2^{|I|}$. Since X_i is non-compact we have $|X_i| > 1$ for $i \in I \setminus J$, and from $|I \setminus J| = |I|$ we have

$$|\{p\} \times X_{I \setminus J}| \geq 2^{|I|},$$

a contradiction.

Case 2. There is a set $\{K_\xi : \xi < \beta\}$ of compact subspaces of X_I with $\beta \leq \alpha$ such that $A \cup A' \subset \bigcup_{\xi < \beta} K_\xi$. We assume without loss of generality that $\beta = \alpha$ and we choose $\{i(\xi) : \xi < \alpha\} \subset I \setminus J$ such that

$$i(\xi') \neq i(\xi) \text{ for } \xi' < \xi < \alpha.$$

There is $x \in X_{I \setminus J}$ such that

$$x_{i(\xi)} \in X_{i(\xi)} \setminus \pi_{i(\xi)}[K_\xi] \quad \text{for } \xi < \alpha,$$

and we have $\langle p, x \rangle \in X_J \times X_{I \setminus J} = X_I$ and $\langle p, x \rangle \notin \bigcup_{\xi < \alpha} K_\xi$; this contradicts the relations

$$\{p\} \times X_{I \setminus J} \subset A \cup A' \subset \bigcup_{\xi < \alpha} K_\xi.$$

The proof that $\pi_J[Y] = X_J$ for $J \in \mathcal{P}_\alpha^*(I)$ is complete.

It follows from Corollary 10.5(ii) that Y is M -embedded in X_I , and since Y is dense in X_I and X_i is metric for $i \in I$, we have

$$\gamma(Y) = \gamma(X_I) = X_I$$

(cf. Theorem B.2). Since $Y \subset X_I \setminus A \subset X_I$, we have $\gamma(X_I \setminus A) = X_I$; that $\gamma(X_I \setminus A') = X_I$ follows similarly.

Now let f be a homeomorphism of $X_I \setminus A$ onto $X_I \setminus A'$. Since

$\gamma(X_I \setminus A) = X_I$ and $\gamma(X_I \setminus A') = X_I$ both f and f^{-1} extend to continuous functions $\bar{f}$ and $\bar{f}^{-1}$ from X_I into X_I . Since

$$(\bar{f} \circ \bar{f}^{-1})(x) = x = (\bar{f}^{-1} \circ \bar{f})(x) \quad \text{for } x \in Y$$

and Y is dense in X_I , we have

$$(\bar{f} \circ \bar{f}^{-1})(x) = x = (\bar{f}^{-1} \circ \bar{f})(x) \quad \text{for } x \in X_I.$$

It follows that $\bar{f}$ and $\bar{f}^{-1}$ are (mutually inverse) homeomorphisms of X_I onto X_I . Since

$$\bar{f}[X_I \setminus A] = f[X_I \setminus A] = X_I \setminus A',$$

we have $\bar{f}[A] = A'$, so that A is homeomorphic to A' .

10.9 Remarks. (a) The assumption in the statement of Theorem 10.8 that the sets K_ξ (with $\xi < \beta$) are compact may be replaced by the weaker assumption that these sets are pseudocompact. To verify this it is sufficient, according to the considerations of Case 2 in the proof of Theorem 10.8, to show that $\pi_{i(\xi)}[K_\xi] \neq X_{i(\xi)}$ for $\xi < \alpha$; for this it is enough to show that every pseudocompact metric space M is compact. Let the indexed family $\{U_n : n < \omega\}$ be a locally finite family of non-empty open subsets of M , and let $x_n \in U_n$ for $n < \omega$. Then the infinite set $\{x_n : n < \omega\}$ has no accumulation point in M . Thus M is not countably compact; it follows that the metric space M is not compact.

(b) We note the following special case of Theorem 10.8 with $\alpha = \omega^+$. Let β be an uncountable cardinal and let p and q be distinct elements of the space $\mathbb{R}^\beta$. Then no two of the three spaces $\mathbb{R}^\beta$, $\mathbb{R}^\beta \setminus \{p\}$, $\mathbb{R}^\beta \setminus \{p, q\}$ are homeomorphic.

(c) The results the Theorem 10.8 are in contrast to the situation in the space $\mathbb{R}^\omega$. Indeed, Anderson (1966) has shown, en route to the important result that $\mathbb{R}^\omega$ is homeomorphic to sequential Hilbert space, that if Y is a compact subset of $\mathbb{R}^\omega$, then $\mathbb{R}^\omega \setminus Y$ is homeomorphic to $\mathbb{R}^\omega$; earlier Klee (1956) had shown that if Y is a compact subspace of an infinite-dimensional normed linear space E , then $E \setminus Y$ and E are homeomorphic.

We continue with two consequences of Corollary 10.7.

10.10 Corollary. Let $\{X_i : i \in I\}$ be a set of spaces, $p \in X_I$ and $\Sigma_{\omega^+}(p, X_I) \subset Y \subset X_I$.

- (a) If $X_i = \mathbb{R}$ for $i \in I$ then $X_I = v(Y)$;
- (b) if X_i is a compact Hausdorff space for $i \in I$, then $X_I = \beta(Y)$.

Definition. Let X be a space, $Y \subset X$ and $x \in X$.

- (a) Y is a G_δ -set of X if there is $\{U_n : n < \omega\} \subset \mathcal{T}^*(X)$ such that $Y = \bigcap_{n < \omega} U_n$;
- (b) x is a G_δ -point if $\{x\}$ is a G_δ -set;
- (c) Y is G_δ -dense in X if $Y \cap U \neq \emptyset$ for every non-empty G_δ -set U of X .

We note that if every element of a space X is a G_δ -point (in particular, if X is metrizable), then for $Y \subset X$ we have: Y is G_δ -dense in X if and only if $Y = X$.

10.11 Corollary. Let $\{X_i : i \in I\}$ be a set of non-empty metric spaces and $Y \subset X_I$.

- (a) If each space X_i is separable, then $X_I = v(Y)$ if and only if Y is G_δ -dense in X_I ;
- (b) if Y is pseudocompact and dense in X_I , then each X_i is compact and $X_I = \beta(Y)$.

Proof. (a) Like every completely regular, Hausdorff space, the space Y is G_δ -dense in $v(Y)$; indeed, otherwise there is $f \in C(v(Y))$ such that $f > 0$ on Y and $f^{-1}(\{0\}) \neq \emptyset$, and then the (continuous) function g defined on Y by the rule $g = 1/(f|Y)$ does not extend continuously over $v(Y)$. If, conversely, Y is G_δ -dense in X_I , then for $J \in \mathcal{P}_\omega^*(I)$ we have $\pi_J[Y]$ G_δ -dense in X_J and hence $\pi_J[Y] = X_J$ (since X_J is metrizable). The (separable, metric) spaces X_i are Lindelöf spaces and hence realcompact, so the required conclusion $X_I = v(Y)$ now follows from Corollary 10.7(a).

(b) For $J \in \mathcal{P}_\omega^*(I)$, the space $\pi_J[Y]$ is pseudocompact and dense in X_J ; since a pseudocompact metrizable space is compact (cf. Remark 10.9(a)), we have $\pi_J[Y] = X_J$, each space X_i is compact, and Corollary 10.7(b) applies.

In 10.12–10.14 below we offer an alternative approach, not dependent on the Erdős–Rado theorem on quasi-disjoint families, to some of the results presented earlier in this chapter.

10.12 Lemma. Let β be a cardinal, $\{X_i : i \in I\}$ a set of spaces, and $A = \bigcup_{\eta < \beta} A_\eta \subset X_I$.

(a) If A depends on $J \subset I$ then $\text{cl } A$ depends on J ; and

(b) if A_η depends on $J_\eta \subset I$ for $\eta < \beta$ then A depends on $\cup_{\eta < \beta} J_\eta$.

Proof. (a) Let $x \in \text{cl } A$, $y \in X_I$ and $x_J = y_J$. There is a net $\langle x(\lambda) : \lambda \in \Lambda \rangle$ of elements of A such that $\lim_\lambda x(\lambda) = x$. For $\lambda \in \Lambda$ we define $y(\lambda) \in X_I$ by

$$\begin{aligned} y(\lambda)_i &= x(\lambda)_i & \text{if } i \in J \\ &= y_i & \text{if } i \in I \setminus J. \end{aligned}$$

Since

$$\begin{aligned} \lim_\lambda y(\lambda)_i &= \lim_\lambda x(\lambda)_i = x_i = y_i & \text{for } i \in J, \text{ and} \\ y(\lambda)_i &= y_i & \text{for } i \in I \setminus J, \end{aligned}$$

we have $\lim_\lambda y(\lambda) = y$; and from $y(\lambda)_J = x(\lambda)_J$ we have $y(\lambda) \in A$. It follows that $y \in \text{cl } A$, as required.

(b) Set $\tilde{J} = \cup_{\eta < \beta} J_\eta$ and let $x \in A$ and $y \in X_I$ with $x_J = y_J$. There is $\tilde{\eta} < \beta$ such that $x \in A_{\tilde{\eta}}$ and from $x_{J_{\tilde{\eta}}} = y_{J_{\tilde{\eta}}}$ we have $y \in A_{\tilde{\eta}} \subset A$, as required.

10.13 Lemma. Let $\alpha \geq \omega$, let $\{X_i : i \in I\}$ be a set of spaces with $d(X_i) \leq \alpha$ for $i \in I$, and U an open subspace of X_I . Then $\text{cl } U$ depends on $\leq \alpha$ coordinates.

Proof. We assume $U \neq \emptyset$. Let $\{U_\eta : \eta < \beta\}$ be a maximal cellular family of basic (in X_I) subsets of U . Since $d(X_i) \leq \alpha$, for $i \in I$ we have

$$S(U) \leq S(X_I) \leq \alpha^+$$

from Theorem 3.28 and hence $\beta \leq \alpha$. From the maximality of the family $\{U_\eta : \eta < \beta\}$ it follows that $\cup_{\eta < \beta} U_\eta$ is dense in U and hence $\text{cl } U = \text{cl}(\cup_{\eta < \beta} U_\eta)$. Since the (basic) set U_η depends on $< \omega$ coordinates, it follows from Lemma 10.12 that $\cup_{\eta < \beta} U_\eta$, and hence its closure, depends on $\leq \alpha$ coordinates.

10.14 Theorem. Let $\alpha \geq \omega$, let $\{X_i : i \in I\}$ be a set of spaces with $d(X_i) \leq \alpha$ for $i \in I$, Y a Hausdorff space with $w(Y) \leq \alpha$ and f a continuous function from X_I into Y . Then f depends on $\leq \alpha$ coordinates.

Proof. Let $\{B_\xi : \xi < \alpha\}$ be a basis for Y , and set $U_\xi = f^{-1}(B_\xi)$ for $\xi < \alpha$. From Lemma 10.13 there is $J_\xi \subset I$ with $|J_\xi| \leq \alpha$ such that $\text{cl } U_\xi$ depends on J_ξ . We set $J = \cup_{\xi < \alpha} J_\xi$, we note that $|J| \leq \alpha$, and we show that f depends on J .

If $x, y \in X_I$ with $x_J = y_J$ and $f(x) \neq f(y)$, then since Y is a Hausdorff space there is $\xi < \alpha$ such that $f(x) \in B_\xi$ and $f(y) \notin \text{cl } B_\xi$, and we have

$$\begin{aligned} x \in f^{-1}(B_\xi) = U_\xi \subset \text{cl } U_\xi, \text{ and} \\ f(y) \notin \text{cl } B_\xi \supset \text{cl } f[U_\xi] \supset f[\text{cl } U_\xi], \end{aligned}$$

and hence $y \notin \text{cl } U_\xi$. This contradiction completes the proof.

The following result is derived from Theorem 10.14 or, alternatively, from Corollary 10.2.

As usual, we say that space X is *normal* if for every pair $\langle A, B \rangle$ of disjoint, closed subsets of X there are disjoint open subsets U and V of X such that $A \subset U$ and $B \subset V$. In the following proof we use Urysohn's lemma: if A and B are disjoint, closed subsets of a normal space X , then there is a continuous function f from X to $[0, 1]$ such that

$$\begin{aligned} f(x) &= 0 \quad \text{for } x \in A, \\ &= 1 \quad \text{for } x \in B. \end{aligned}$$

10.15 *Theorem.* (a) The space $\omega^{(\omega^+)}$ is not normal.

(b) If $\{X_i : i \in I\}$ is a set of non-empty spaces with X_I normal, then

$$|\{i \in I : X_i \text{ is not countably compact}\}| \leq \omega.$$

Proof. (a) For $k = 0, 1$ we set

$$A_k = \{x \in \omega^{(\omega^+)} : \text{if } n \neq k \text{ then } |\{\xi < \omega^+ : x_\xi = n\}| \leq 1\}.$$

It is clear that A_0, A_1 are disjoint, closed subsets of $\omega^{(\omega^+)}$; hence if $\omega^{(\omega^+)}$ is normal there is $f \in C(\omega^{(\omega^+)}, [0, 1])$ such that

$$\begin{aligned} f(x) &= 0 \quad \text{for } x \in A_0, \\ &= 1 \quad \text{for } x \in A_1. \end{aligned}$$

From Theorem 10.14 (with $\alpha = \omega$) or Corollary 10.2 (with $\kappa = \omega$ and $\alpha = \omega^+$) there is $J \subset \omega^+$ with $|J| = \omega$ such that f depends on J ; we write $J = \langle \xi_n : n < \omega \rangle$ with

$$\xi_{n'} \neq \xi_n \quad \text{for } n' < n < \omega.$$

We define $p, q \in \omega^{(\omega^+)}$ by

$$\begin{aligned} p_\xi &= q_\xi = n \quad \text{for } \xi = \xi_n \in J \\ p_\xi &= 0 \quad \text{for } \xi \in \omega^+ \setminus J, \\ q_\xi &= 1 \quad \text{for } \xi \in \omega^+ \setminus J. \end{aligned}$$

Since $p \in A_0$ and $q \in A_1$, we have $f(p) = 0$ and $f(q) = 1$, and since $p_J = q_J$, we have $f(p) = f(q)$. This contradiction completes the proof.

(b) If the result fails there is $J \subset I$ with $|J| = \omega^+$ such that X_i is not countably compact for $i \in J$. The normal space X_J then contains a closed subspace homeomorphic to the non-normal space ω^J , a contradiction.

We conclude this chapter with three miscellaneous results which answer questions raised naturally by what has appeared above. The first is a partial converse to Corollary 10.2; the second shows that an ω^+ - Σ -product in a product of first-countable spaces is C -embedded, thus substantially generalizing an important special case of Theorem 10.4; and the last shows that not every ω^+ - Σ -product is C -embedded.

10.16 Theorem. Let $\alpha > \omega$ with α regular and let $\{X_i : i \in I\}$ be a set of completely regular, Hausdorff spaces with $|I| \geq \alpha$ and $|X_i| \geq 2$ for $i \in I$. If every continuous function from X_I to $[0, 1]$ depends on $< \alpha$ coordinates, then X_I is pseudo- α -compact.

Proof. If the statement fails, then by Theorem 3.6(c) there is $J \in \mathcal{P}_\omega^*(I)$ such that X_J is not pseudo- α -compact. Let $\{U_\xi : \xi < \alpha\}$ be a locally finite, cellular family in X_J , let $\{i_\xi : \xi < \alpha\}$ be a faithfully indexed subset of $I \setminus J$, let V_ξ be an open subset of X_{i_ξ} for $\xi < \alpha$ with $V_\xi \neq X_{i_\xi}$, and set

$$W_\xi = U_\xi \times V_\xi \times X_{I \setminus (J \cup \{i_\xi\})} \quad \text{for } \xi < \alpha.$$

The (indexed) family $\{W_\xi : \xi < \alpha\}$ is a locally finite, cellular family in X_I . For $\xi < \alpha$ we choose $x(\xi) \in W_\xi$ and $f_\xi \in C(X_I, [0, 1])$ such that

$$\begin{aligned} f_\xi(x(\xi)) &= 1, \quad \text{and} \\ f_\xi(x) &= 0 \quad \text{for } x \in X_I \setminus W_\xi, \end{aligned}$$

and we set $f = \sum_{\xi < \alpha} f_\xi$. Since

$$W_{\xi'} \cap W_\xi = \emptyset \quad \text{for } \xi' < \xi < \alpha,$$

we have $f[X_I] \subset [0, 1]$, and since $\{W_\xi : \xi < \alpha\}$ is locally finite and $f_\xi(x) = 0$ for $x \notin W_\xi$, the function f is continuous.

To complete the proof it is enough to show that if $J' \subset I$ and f depends on J' , then $\{i_\xi : \xi < \alpha\} \subset J'$. We fix $\xi < \alpha$. Since $V_\xi \subsetneq X_{i_\xi}$,

there is $y(\xi) \in X_I$ such that

$$y(\xi)_i = x(\xi)_i \quad \text{for } i \neq i_\xi$$

$$y(\xi)_{i_\xi} \in X_{i_\xi} \setminus V_\xi.$$

Now for $\eta < \alpha$ with $\eta \neq \xi$ we have $x(\xi) \notin W_\eta$ and also

$$y(\xi)_J = x(\xi)_J \in U_\xi \setminus U_\eta,$$

and hence $y(\xi) \notin W_\eta$; it follows that $f_\eta(x(\xi)) = 0$ and $f_\eta(y(\xi)) = 0$ for $\eta \neq \xi$, and from $y(\xi)_{i_\xi} \notin V_\xi$ (and hence $y(\xi) \notin W_\xi$) we have

$$f(x(\xi)) = f_\xi(x(\xi)) = 1, \quad \text{and}$$

$$f(y(\xi)) = f_\xi(y(\xi)) = 0.$$

It is clear that if f depends on J' then $i_\xi \in J'$, as required.

A space X is *first-countable* if for $x \in X$ there is a countable neighborhood base at x .

10.17 Theorem. Let $\{X_i : i \in I\}$ be a set of first-countable spaces and $p \in X_I$. Then $\Sigma_{\omega^+}(p)$ is C -embedded in X_I .

Proof. If the statement fails there are $f \in C(\Sigma_{\omega^+}(p))$ and $q \in X_I \setminus \Sigma_{\omega^+}(p)$ such that f does not extend continuously to q ; since $\mathbb{R}$ is complete there is $\varepsilon > 0$ such that the oscillation of f is at least ε on every neighborhood in X_I of q . For U a neighborhood of q in X_I we choose $x_U, y_U \in U \cap \Sigma_{\omega^+}(p)$ such that

$$|f(x_U) - f(y_U)| > \varepsilon;$$

we assume without loss of generality, since f is continuous and $\Sigma_{\omega}(p)$ is a dense subspace of $\Sigma_{\omega^+}(p)$, that $x_U, y_U \in \Sigma_{\omega}(p)$.

Now for $i \in I$ and $z \in X_i$ let $\{U_n(z) : n < \omega\}$ be a neighborhood base at z with $U_{n+1}(z) \subset U_n(z)$ for $n < \omega$. (To avoid ambiguity in this choice we assume without loss of generality that $X_{i'} \cap X_i = \emptyset$ for $i, i' \in I$ with $i \neq i'$.) For $n < \omega$ we define $J(n)$, $U(n)$, $x(n)$, $y(n)$ as follows:

- (i) $J(0) = \emptyset$;
- (ii) $U(0) = X_I$;
- (iii) $x(0) = x_{U(0)}$, $y(0) = y_{U(0)}$;
- (iv) $J(n+1) = J(n) \cup \{i \in I : x(n)_i \neq p_i \text{ or } y(n)_i \neq p_i\}$ for $n < \omega$;
- (v) $U(n+1) = \{z \in X_I : z_i \in U_{n+1}(q_i) \text{ for } i \in J(n)\}$ for $n < \omega$; and
- (vi) $x(n+1) = x_{U(n+1)}$, $y(n+1) = y_{U(n+1)}$ for $n < \omega$.

We define $\bar{z} \in X_I$ by

$$\begin{aligned}\bar{z}_i &= q_i \quad \text{for } i \in \bigcup_{n < \omega} J(n) \\ &= p_i \quad \text{for } i \in I \setminus \bigcup_{n < \omega} J(n)\end{aligned}$$

Since $|J(n)| < \omega$ for $n < \omega$ we have $\bar{z} \in \Sigma_{\omega^+}(p)$. We claim further that

$$\lim_n x(n) = \lim_n y(n) = \bar{z}.$$

Indeed, if $i \in J(k)$ with $k < \omega$, then for $k < m < n < \omega$ we have

$$\begin{aligned}x(n)_i &\in U_n(q_i) \subset U_m(q_i) = U_m(\bar{z}_i), \quad \text{and} \\ y(n)_i &\in U_n(q_i) \subset U_m(q_i) = U_m(\bar{z}_i)\end{aligned}$$

and hence

$$\lim_n x(n)_i = \lim_n y(n)_i = \bar{z}_i,$$

and if $i \notin \bigcup_{n < \omega} J(n)$, then

$$x(n)_i = y(n)_i = p_i = \bar{z}_i$$

and hence

$$\lim_n x(n)_i = \lim_n y(n)_i = \bar{z}_i.$$

The claim is established. Since

$$|f(x(n)) - f(y(n))| > \varepsilon \quad \text{for } n < \omega,$$

the function f is not continuous at the point $\bar{z}$. This contradiction completes the proof.

It follows from Theorem 10.17 that in a product of metrizable spaces, and in particular in a product of discrete spaces, every ω^+ - Σ -product is C -embedded. Together with the results of Corollary 10.5, this raises the question whether every ω^+ - Σ -product is C -embedded. In this connection we note a curiosity. If $f \in C(\Sigma_{\omega^+}(p))$ with $p \in X_I$ and if $q \in X_I \setminus \Sigma_{\omega^+}(p)$, there is a natural way to attempt to extend f continuously to q : give the spaces X_i the discrete topology, note that f remains continuous on $\Sigma_{\omega^+}(p)$ in the stronger topology now inherited from the (new) product space X_I , and extend f to q using Theorem 10.17. It is clear that if relative to the original product topology the function f extends continuously to q , it is with the value just obtained; and the question of determining whether a function continuous on an ω^+ - Σ -product extends continuously

over X_I becomes the question whether a particular function, known to be continuous relative to a certain strong topology, is continuous relative to the usual product topology.

We show now that not every ω^+ - Σ -product is C -embedded.

10.18 Theorem. Let $\alpha \geq \omega$. There are a set $\{X_i : i \in I\}$ of completely regular, Hausdorff spaces, $p, q \in X_I$ and $f \in C(\Sigma_\alpha(p), \{0, 1\})$ such that no continuous function from $\Sigma_\alpha(p) \cup \{q\}$ to $[0, 1]$ extends f .

Proof. We note first that there is a family $\mathcal{F}$ of subsets of α such that

- (i) $|\mathcal{F}| = \alpha$;
- (ii) $|A| = \alpha$ for A in $\mathcal{F}$;
- (iii) if $\mathcal{G} \subset \mathcal{F}$ and $|\mathcal{G}| < \omega$, then $\bigcap \mathcal{G} \neq \emptyset$; and
- (iv) if $\mathcal{G} \subset \mathcal{F}$ and $|\mathcal{G}| \geq \omega$, then $\bigcap \mathcal{G} = \emptyset$.

(To define such a family $\mathcal{F}$, let $\{B_\xi : \xi < \alpha\}$ be a faithful indexing of $\mathcal{P}_\omega^*(\alpha)$, and set $A_\zeta = \{\xi < \alpha : \xi \in B_\xi\}$ for $\zeta < \alpha$ and $\mathcal{F} = \{A_\zeta : \zeta < \alpha\}$.) Let $q_0 \in \beta(\alpha)$ with $\mathcal{F} \subset q_0$ and set $X_0 = \alpha \cup \{q_0\}$ with the topology inherited from $\beta(\alpha)$.

For $A \in \mathcal{F}$ let X_A denote the one-point compactification $A \cup \{q_A\}$ of the (discrete) space A , set

$$X = X_0 \times \prod_{A \in \mathcal{F}} X_A,$$

and let $p \in X$ satisfy

$$\begin{aligned} p_0 &\neq q_0, \\ p_A &\neq q_A \text{ for } A \in \mathcal{F} \end{aligned}$$

Now we set

$$K = \{x \in \Sigma_\alpha(p) : x_0 \neq q_0, \text{ and } x_A = x_0 \text{ whenever } x_0 \in A \in \mathcal{F}\}.$$

Because no element of α lies in infinitely many elements of $\mathcal{F}$, the set K is open in $\Sigma_\alpha(p)$. (In detail: if $x \in K$ and $x_0 = \xi \in \alpha$, then

$$\pi_0^{-1}(\{\xi\}) \cap \bigcap \{\pi_A^{-1}(\{x_A\}) : \xi \in A \in \mathcal{F}\},$$

is a neighborhood in X of x whose intersection with $\Sigma_\alpha(p)$ is a subset of K .)

We claim also that K is closed in $\Sigma_\alpha(p)$. Let $x \in \Sigma_\alpha(p) \setminus K$. If $x_0 \neq q_0$,

there is $A \in \mathcal{F}$ such that $x_A \neq x_0$ and then

$$\pi_0^{-1}(\{x_0\}) \cap \pi_A^{-1}(\{x_A\})$$

is a neighborhood in X of x disjoint from K . If $x_0 = q_0$, then since $|\mathcal{F}| = \alpha$ and $x \in \Sigma_\alpha(p)$ there is $A \in \mathcal{F}$ such that $x_A = p_A$ and there is $B \in \mathcal{F}$ such that $x_A \in \alpha \setminus B$; we set

$$U = \{q_0\} \cup (A \cap B).$$

Then U is a neighborhood in X_0 of q_0 and hence

$$\pi_0^{-1}(U) \cap \pi_A^{-1}(\{x_A\})$$

is a neighborhood in X of x ; this misses K , for if $y_0 \in U$ then either $y_0 \neq q_0$ (hence $y \notin K$) or $y_0 \in A \cap B$ (hence $y_A \neq x_A$). The proof that K is closed in $\Sigma_\alpha(p)$ is complete.

To complete the proof it is enough to show that every neighborhood in X of q meets both K and $\Sigma_\alpha(p) \setminus K$, for then the function which is 1 on K and 0 on $\Sigma_\alpha(p) \setminus K$ will admit no continuous extension to q . Let $V = V_0 \times \prod_{A \in \mathcal{F}} V_A$ be a neighborhood in X of q with (finite) restriction set $R(V)$, and set

$$\begin{aligned} y_0 &= q_0 \\ y_A &= q_A \quad \text{for } A \in R(V) \\ &= p_A \quad \text{for } A \in \mathcal{F} \setminus R(V); \end{aligned}$$

then $y \in V \cap (\Sigma_\alpha(p) \setminus K)$.

We define $z \in V \cap K$. Since $|A \setminus V_A| < \omega$ for $A \in \mathcal{F}$, we have $V_A \in q_0$ for $A \in \mathcal{F} \subset q_0$; hence there is

$$z_0 \in V_0 \cap \bigcap \{V_A : A \in R(V) \cap \mathcal{F}\}.$$

We set $\mathcal{F}' = \{A \in \mathcal{F} : z_0 \in A\}$, we note that $\mathcal{F}'$ is a finite subset of $\mathcal{F}$ such that $R(V) \cap \mathcal{F} \subset \mathcal{F}'$, and we set

$$\begin{aligned} z_A &= z_0 \quad \text{for } A \in \mathcal{F}' \\ &= p_A \quad \text{for } A \in \mathcal{F} \setminus \mathcal{F}'. \end{aligned}$$

It is then clear that $z \in V \cap K$.

The proof is complete.

Notes for Chapter 10

From Corollary 10.2 it follows that under certain conditions every continuous function from a product space X_I to a metric space

depends on countably many coordinates. When each X_i is compact, this is due to Mibu (1944) and Bishop (1959); when each is separable and metrizable, to Mazur (1952) (with an additional cardinality hypothesis, but for sequentially continuous functions) and Corson & Isbell (1960); when each has property (K), to Ross & Stone (1964); when each is separable, to Gleason (see Ross & Stone (1964) or Isbell (1964, pp. 130–2)); when each finite product of the X_i is a Lindelöf space, to Engelking (1966); and when X (or each X_F with $F \in \mathcal{P}_\omega^*(I)$) is pseudo- ω^+ -compact, to Noble & Ulmer (1972). Using his own generalization of the concept of calibre as introduced by Shanin (1946b), Miščenko (1966a, b) has given analogous statements in the context of uniform spaces (for uniformly continuous functions). Some of these authors consider functions defined on subspaces of X_J , and some impose hypotheses weaker than metrizability on the range space.

That every continuous function from a product of compact, Hausdorff spaces to the real line or to the complex plane depends on countably many coordinates is a direct consequence of the Stone–Weierstrass theorem noted by many authors: the algebra of continuous functions depending on finitely many coordinates separates points and distinguishes points from closed sets (and is conjugate-closed), and clearly every function in its uniform closure depends on $\leq \omega$ coordinates. For a related argument using partitions of unity, for functions defined on the product of two compact spaces, see Dieudonné (1937).

In addition to the new contributions they contain, the works of Engelking (1966) and Hušek (1976) are valuable surveys of the literature of continuous functions defined on product spaces and their subspaces; Hušek (1976) has classified into six groupings the patterns of proof used by the authors he cites.

The fact that continuous functions from a product to spaces subjected to a countability condition may fail to depend on countably many coordinates has been noted by Miščenko (1966a), Engelking (1968, p. 20), and Marty (1969, Example 20). The argument we use in Theorem 10.16 is suggested by Noble & Ulmer (1972, Theorem 2.3).

The argument of 10.12–10.14 was introduced by Bockstein (1948)

to show that disjoint, open subsets in a product of separable metric spaces remain disjoint when projected into a suitable countable subproduct; see also Corson & Isbell (1960). In our treatment of 10.12–10.16 we have followed Ross & Stone (1964); Theorem 10.15, first proved by Stone (1948), has been generalized by Noble (1967).

It should be noted that statements weaker than Theorem 10.15 can be proved easily and directly. To see for example that $\mathbb{R}^{2^\omega}$ is not normal, let X be a separable, realcompact space with a closed discrete subspace D such that $|D| = 2^\omega$ (e.g., let $X = S \times S$, with S the ‘Sorgenfrey line’ as described by Sorgenfrey (1947) or Kelley (1955, Problem 4I)), and embed X into $\mathbb{R}^{C(X)}$ using the function $e: X \rightarrow \mathbb{R}^{C(X)}$ defined by $e(x)_f = f(x)$. Since $e[X]$, which is homeomorphic to X , is realcompact and C -embedded in $\mathbb{R}^{C(X)}$, the space $e[X]$ is closed in $\mathbb{R}^{C(X)}$ (cf. Theorem B.4 of the Appendix). Since $|C(D)| = 2^{2^\omega}$ and $|C(X)| = 2^\omega$, it follows from Tietze’s extension theorem that X is not normal; hence $\mathbb{R}^{C(X)}$, which is $\mathbb{R}^{2^\omega}$, is not normal.

The recursive argument of Lemma 10.1 was introduced by Glicksberg (1959) to prove the case $\kappa = \alpha = \omega$; see also Kister (1962). Noble & Ulmer (1972) proved Lemma 10.1 and Corollary 10.2(b) in the case that $\kappa = \omega$ and Y contains an ω^+ - Σ -product. The present treatment of 10.1–10.4 follows closely Comfort & Negreponitis (1972b).

ω^+ - Σ -products were introduced and systematically studied by Corson (1959), who established the relation $v(\Sigma_{\omega^+}(p, X_I)) = X_I$ for X_i separable metric; Corollary 10.7(b) (for ω^+ - Σ -products) was given independently by Glicksberg (1959) and Kister (1962). Parts (a) and (b) of Corollary 10.11 are from Efimov & Engelking (1965), and Noble (1972), respectively.

Van der Slot (1972) showed that two compact subspaces of $\mathbb{R}^\alpha$ (with $\alpha > \omega$) are homeomorphic if and only if their complements are homeomorphic. Our proof of Theorem 10.8 follows arguments suggested by Hušek (1972, Corollary 3) and Hušek (1976, (5), p. 39).

Theorems 10.17 and 10.18 are from the thesis of Ulmer (1970), which contains several results more general than those given here. For example: every ω^+ - Σ -product in a product of P_{ω^+} -spaces is C -embedded, but there are spaces $\{X_i : i \in I\}$, all but one a P_{ω^+} -space

and each containing exactly one non-isolated point, such that some ω^+ - Σ -product in X_I is not C -embedded.

We note that (as is mentioned in the Notes for Chapter 7) the class of Eberlein-compact spaces is contained in the class of Corson-compact spaces. The preservation of Eberlein-compact spaces under continuous functions is a basic result that has been extended to various other classes of subspaces of Σ -products. See in this connection Benyamini, Rudin & Wage (1977), Michael & Rudin (1977), Gul'ko (1977, 1978), Alster & Pol (1980), and Negrepontis & Tsarpalias (1981).